APOLLOLINK (MEDICAL ASSISTANCE ROVER)

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Abstract: This project presents the design and development of an autonomous medical rover equipped with advanced technologies such as RF beacon(LoRa sx1278) navigation, LiDAR(TF-Luna), RFID/NFC modules, heart rate monitoring(ky-039), and a smart medicine box. The rover is designed to autonomously navigate healthcare facilities while monitoring patient vitals and dispensing medication. Its modular design ensures ease of use and adaptability to various healthcare scenarios.

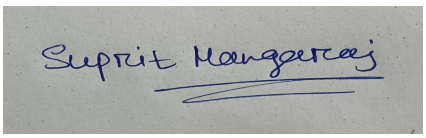
Individual contribution and findings: I primarily worked on integrating the LoRa SX1278 modules for rover navigation, ensuring seamless communication between the transmitter at the destination and the receiver on the rover. One major challenge I encountered was that while the transmitter was successfully sending signals, the receiver was not receiving them. I troubleshot this issue by checking wiring connections, verifying SPI communication, adjusting frequencies, and debugging the receiver code. After multiple tests and refinements, I resolved the issue by properly configuring the LoRa library, optimizing signal strength, and minimizing interference. This process enhanced my understanding of wireless communication, debugging techniques, and real-time signal processing.

Individual contribution to project report preparation: I contributed to the project report by documenting the integration of LoRa SX1278 for navigation and RFID for authentication. I explained how the LoRa receiver on the rover used signal strength to determine movement and the challenges faced in receiving data reliably. I also detailed the RFID authentication process, ensuring secure access to the system. Additionally, I worked on troubleshooting LoRa communication issues and optimizing RFID response, ensuring smooth system functionality.

Individual contribution for project presentation and demonstration: For the project presentation and demonstration, I will explain the integration of LoRa SX1278 for navigation and RFID for authentication. I will demonstrate how the rover follows the LoRa signal to reach its destination and how RFID is used for secure access. I will also highlight the challenges faced in LoRa communication, especially in data reception, and explain how we optimized the system for better performance. Additionally, I will showcase the obstacle detection mechanism and ensure a clear understanding of the project's functionality during the demonstration.

Full Signature of Supervisor/s:

Full signature of the student:



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